Computational Statistics

9 November 2026

Dear Editors,

Please consider our manuscript, "HGFX: reference-compatible hierarchical Gaussian filtering in Python," as a software contribution to Computational Statistics. HGFX addresses a practical statistical-computing problem: moving established Hierarchical Gaussian Filter analyses from MATLAB into Python while preserving the tested meaning of model configurations, parameter transformations, priors, trial masks and fitted outputs. The contribution is a released research package and an auditable evaluation of its compatibility boundaries, rather than a new statistical model or a claim of improved neuroscientific inference.

The manuscript connects the analysis workflow to a frozen MATLAB HGF Toolbox 8.2.0 reference. It separates configuration and trajectory checks from fitting, model selection and recovery. In the frozen paired experiment, all 36 Bayesian information criterion winners agree across implementations, while generating-model recovery balanced accuracy is 0.583 in each. Failed parameter-recovery criteria and the failed integrity gate of a separate trial-horizon experiment remain explicit. A comparison with pyhgf is restricted to one prospectively mapped perceptual case; response negative log-likelihood remains not directly comparable.

The software and accompanying source, evidence manifests and reproducibility instructions are publicly available through the HGFX repository. A released-wheel example applies the continuous HGF to all 614 observations of the reference toolbox's empirical USD/CHF series. The evaluation also compares fixed-parameter perceptual outputs across MATLAB and HGFX on 324 source-defined blocks from two published behavioral input files. All blocks met the prespecified numerical criterion. The paper reports the warnings and distinguishes this compatibility result from fitting or reproducing either study's scientific conclusions.

We believe the combination of statistical software, numerical evidence and a clear account of inferential limitations fits the journal's stated package-report and software-article scope. A related preprint is posted under DOI 10.64898/2026.09.22.753415; this draft is a proposed revision and has not been uploaded. We intend the subscription route, subject to confirmation that it incurs no mandatory author charges. Authorship and declarations have been confirmed. This letter accompanies the author-review package; final scientific and administrative approval remain pending.

Sincerely,

Mohammad Ahmadkhanloo